

6.6 Equation of a Line Given Two Points

Remember: You can write the equation of a line once you know the **slope** and **y-intercept**.

$$y = mx + b$$

The diagram illustrates the components of the line equation $y = mx + b$. A box labeled "Slope" has an arrow pointing to the variable m . Another box labeled "y-intercept" has an arrow pointing to the variable b .

DO IT NOW!

Instructions: Write the equation of the following lines:

- a) Line with a slope of $\frac{3}{5}$ that passes through the point B(-5, 4).

$$y = mx + b$$

$$4 = \left(\frac{3}{5}\right)(-5) + b$$

$$4 = -\frac{15}{5} + b$$

$$4 = -3 + b$$

$$4 + 3 = b$$

$$b = 7$$

$$y = \frac{3}{5}x + 7$$

- b) Line that is parallel to the line $y = 2x - 7$ and passes through the point (1, -3).

$$m = 2$$

$$y = mx + b$$

$$-3 = 2(1) + b$$

$$-3 = 2 + b$$

$$-3 - 2 = b$$

$$b = -5$$

$$y = 2x - 5$$

c) Line that is perpendicular to the line $2x - 2y + 4 = 0$ and passes through the point $(-2, 5)$.

$$m = -1$$

Point: $(-2, 5)$

$$y = mx + b$$

$$5 = (-1)(-2) + b$$

$$5 = 2 + b$$

$$5 - 2 = b$$

$$b = 3$$

$$y = -1x + 3$$



$$-2y = -2x - 4$$

$$y = \frac{-2}{-2}x - \frac{4}{-2}$$

$$y = 1x + 2$$

$$m = 1$$

Today's Lesson: Find the equation of a line given two points on the line.

What do you need to write the equation of a line?

slope (m) and y-intercept (b)

If you are not given the slope of a line, how can you find it?

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

How can you find the y-intercept?

use the slope (m) and the coordinates (x, y) of any point on the line to solve for the y-intercept (b).

Example 1: Determine the equation of a line that passes through the points M(4, -3) and N(2, 5).
 x_1 y_1 x_2 y_2

Step 1: Calculate the slope

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{5 - (-3)}{2 - 4} = \frac{8}{-2} = -4$$

Step 2: Find the y-intercept

$$\begin{aligned} y &= mx + b \\ 5 &= (-4)(2) + b \\ 5 &= -8 + b \\ 5 + 8 &= b \\ b &= 13 \end{aligned}$$

Note: to find the y-intercept you can use any point that is on the line for your x and y values.

Step 3: Write the equation of the line

$$y = -4x + 13$$

Example 2: Determine the equation of a line that passes through the points P(0, 4) and Q(7, 0).

x_1 y_1 x_2 y_2

Step 1: Calculate the slope

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{0 - 4}{7 - 0} \\ &= -\frac{4}{7} \end{aligned}$$

Step 2: Find the y-intercept

$$\begin{aligned} y &= mx + b \\ 4 &= \left(-\frac{4}{7}\right)(0) + b \\ 4 &= b \end{aligned}$$

Step 3: Write the equation of the line

$$y = -\frac{4}{7}x + 4$$

Example 3: Determine the equation of a line that passes through the points A(-4, 2) and B(8, 11).

x_1 y_1 x_2 y_2

Step 1: Calculate the slope

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{11 - 2}{8 - (-4)} \\ &= \frac{9}{12} \\ &= \frac{3}{4} \end{aligned}$$

Step 2: Find the y-intercept

$$\begin{aligned} y &= mx + b \\ 2 &= \left(\frac{3}{4}\right)(-4) + b \\ 2 &= -\frac{12}{4} + b \\ 2 &= -3 + b \\ 2 + 3 &= b \\ b &= 5 \end{aligned}$$

Step 3: Write the equation of the line

$$y = \frac{3}{4}x + 5$$

Example 4: On your own determine the equation of the line that passes through the points A(2,-4) and B(5,5)

x_1 y_1 x_2 y_2

SLOPE

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{5 - (-4)}{5 - 2} \\ &= \frac{9}{3} \\ &= 3 \end{aligned}$$

Y-int

$$\begin{aligned} y &= mx + b \\ 5 &= 3(5) + b \\ 5 &= 15 + b \\ 5 - 15 &= b \\ b &= -10 \end{aligned}$$

$$y = 3x - 10$$

Example 5:

a) An appliance repair company charges \$205 for a repair that takes 3 hours. The same company charges \$505 for a repair that takes 8 hours. Determine an equation that represents the cost of a repair based on the number of hours that the repair takes. y

Hint: you can write two coordinate points with the information given.
(ind. variable, dep. variable)

Point 1: (x_1, y_1) (3, 205)

Point 2: (x_2, y_2) (8, 505)

Slope

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$
$$= \frac{505 - 205}{8 - 3}$$
$$= \frac{300}{5}$$
$$= 60$$

y-int

$$y = mx + b$$
$$205 = 60(3) + b$$
$$205 = 180 + b$$
$$205 - 180 = b$$
$$b = 25$$

$$y = 60x + 25 \quad \text{OR} \quad \text{cost} = 60(\text{hours}) + 25$$

b) What is the cost of a repair that takes 7 hours? y x

$$y = 60x + 25$$

$$y = 60(7) + 25$$

$$y = 420 + 25$$

$$y = 445$$

\$445

c) If a repair costs \$385, how many hours does it take? y x

$$y = 60x + 25$$

$$385 = 60x + 25$$

$$385 - 25 = 60x$$

$$\frac{360}{60} = \frac{60x}{60}$$

$$\frac{360}{60} = x$$

$$x = 6$$

6 hours

Consolidate:

To write the equation of a line you need the slope and y-intercept.

If you are not given the slope you can find it if you have 2 points on the line by using the

Formula: $m = \frac{y_2 - y_1}{x_2 - x_1}$